3D UAV Trajectory Planning With Obstacle Avoidance for UAV-Enabled Time-Constrained Data Collection Systems

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***Abstract*—This paper investigates the three-dimensional (3D) unmanned aerial vehicle (UAV) trajectory-planning problem in a UAV-enabled data collection (DC) system, in which a UAV executes a mission to collect time-constrained data in a geographical area with spatial obstacles. For such a UAV trajectory-planning problem, how to timely collect data, avoid collisions with spatial obstacles, and complete DC with limited UAV onboard energy is a big challenge. To address this challenge, the investigated problem is formulated as an optimization problem aiming to minimize the UAV's total mission completion time subject to the time constraint of collected data, the avoidance of spatial obstacles, and the energy constraint of the UAV. To solve the formulated problem, the problem is decomposed into a visiting sequence optimization sub-problem, and a hovering position and flying path optimization sub-problem. Correspondingly, a genetic algorithm (GA) based algorithm, and a successive convex approximation (SCA) based algorithm are proposed to solve the sub-problems, respectively. Based on the two algorithms, a 3D trajectory optimization algorithm is further proposed to solve the main problem. Simulation results show that the UAV's trajectory planned with the proposed algorithm can effectively reduce the UAV's total mission completion time.**

***Index Terms*—UAV, data collection, obstacle avoidance, 3D trajectory planning.**

I. Introduction

UNMANNED aerial vehicles (UAVs) have been widely used in more and more applications [1], [2]. In many applications, UAV can be deployed to execute a data collection (DC) mission in a geographical area to collect data sensed by devices distributed in the area or to sense and collect data directly in different data collection areas (DCAs) using onboard sensors to provide a variety of DC applications [3]. In executing a DC mission, a UAV needs to fly between different DCAs. The planning of the UAV's trajectory is critical for successfully completing the DC mission. In many DC application scenarios, collected data are often time-constrained and thus need to be timely transmitted to a ground center through a base station (BS) within a limited time [4]. However, if a UAV is far away from a BS, the signal-to-noise ratio (SNR) at the BS may not be able to reach a minimum level, resulting in the UAV unable to transmit collected data directly to the BS. In this case, the UAV has to fly to a position closer to the BS for transmission. This would cause a certain delay and thus affect the timely transmission of collected data. Thus, the planning of the UAV's trajectory should consider the time constraint of collected data. On the other hand, when a UAV flies between different DCAs, it may encounter various spatial obstacles such as mountain peaks, signal towers, or high-rising buildings [5]. To avoid collisions with such obstacles, a common way is to make a UAV fly at a constant and safe altitude, which would result in a two-dimensional (2D) UAV trajectory, and thus limit the sufficient utilization of a UAV's high maneuverability in space and degrade the overall system performance. Thus, it is necessary to consider obstacle avoidance in a three-dimensional (3D) UAV trajectory planning. In addition, since the onboard energy of a UAV is limited, the energy constraint of the UAV is also a concern in the planning of the UAV's trajectory. Therefore, it is necessary to consider all the above issues in the planning of a UAV's trajectory. However, to the best of our knowledge, no previous work has studied the 3D UAV trajectory-planning problem in a UAV-enabled DC system, while considering the time constraint of collected data, the avoidance of obstacles, and a UAV's energy constraint, which is the motivation of this paper.

In this paper, we investigate the 3D UAV trajectory-planning problem in a UAV-enabled DC system, in which a UAV is deployed to execute a mission to collect time-constrained data in a number of DCAs in a geographical area with spatial obstacles. The problem is formulated as an optimization problem with an objective to minimize the UAV's total mission completion time, taking into account the time constraint of collected data, the avoidance of the obstacles, and the UAV's energy constraint. To solve the formulated problem, the problem is decomposed into a visiting sequence optimization sub-problem, and a hovering position and flying path optimization sub-problem. Correspondingly, a genetic algorithm (GA) based algorithm, and a successive convex approximation (SCA) based algorithm are proposed to solve the two sub-problems, respectively. Based on the two algorithms, a 3D trajectory optimization algorithm is

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further proposed to solve the main problem. Simulation results are shown to evaluate the performance of the proposed algorithm in reducing the UAV's total mission completion time.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the system model and problem formulation. Section IV presents the proposed algorithm. Section V conducts performance evaluation. Section VI concludes this paper.

II. Related Work

The planning of a UAV's trajectory is an important problem in UAV-enabled DC systems and has been extensively studied in literature [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17]. In [6], You et al. studied a 3D UAV trajectory-planning problem for maximizing the minimum DC rate from all senor nodes (SNs), and proposed a hybrid offline-online optimization method leveraging both statistical and real-time channel state information (CSI). In [7], Wang et al. considered a UAV-assisted internet of thing system in a sophisticated 3D environment with imperfect CSI, and studied a 3D UAV trajectory-planning problem for minimizing the total mission completion time, while taking into account throughput and flight movement constraints. In [8], Li et al. studied a UAV trajectory-planning problem aiming to minimize the total mission completion time by jointly optimizing the UAV's flying and collection trajectory and proposed a V-shaped collection trajectory to reduce a UAV's collection time. In [9], Chen et al. considered a multi-UAV assisted DC system, and studied a multi-UAV trajectory-planning problem for maximizing the total throughput and energy efficiency of the system, while taking into account the signal-to-interference-and-noise ratio and fairness among heterogeneous ground users. In [10], Yuan et al. studied the UAV trajectory-planning problem in a UAV-assisted DC system, aiming to minimize the total mission completion time by jointly optimizing the UAV trajectory and the SN assignment. In [11], Nguyen et al. studied a 3D UAV trajectory-planning problem aiming to maximize collected data by using a deep reinforcement learning based technique. In [12], Sun et al. considered a system where UAV is dispatched to harvest solar energy and collect data from multiple smart devices (SDs), and studied a 3D UAV trajectory-planning problem aiming to achieve max-min fairness among SDs' uploading data. However, the aforementioned studies considered 2D or 3D scenarios with only delay-tolerant data. The solutions obtained in these studies are inapplicable in scenarios with time-constrained data.

For the UAV trajectory-planning problem in time-constrained DC, many studies have also been conducted. For example, Zhang et al. considered a cellular-enabled UAV DC and data transmission (DT) system, and studied a UAV trajectory-planning problem for minimizing the AoI of collected data with a proposed sensing and transmission trade-off design [4]. In [13], Samir et al. considered a time-constrained DC scenario in an Internet of Things (IoT) network, and studied a UAV trajectory-planning problem for maximizing the number of served devices, taking into account collected data's minimum amount and deadline constraints. In [14], we considered a UAV-enabled DC scenario, and studied a UAV trajectory-planning problem for minimizing the UAV's total mission completion time with the time constraint of collected data and the energy constraint of the UAV. In [15], Li et al. considered a DC scenario in an IoT network, and studied a UAV trajectory-planning problem for minimizing the age of information (AoI) of collected data with no knowledge of the sampling mode used by each SN. In [16], Al-Hilo et al. considered a time-constrained DC scenario in an IoT network, which uses a reconfigurable intelligent surface to improve the UAV's connectivity and energy efficiency, and studied a UAV trajectory-planning problem for maximizing the number of served devices. In [17], Gao et al. studied a multi-UAV trajectory-planning problem for minimizing SNs' peak and average AoI in wireless sensor networks. Nevertheless, most of the above studies considered a constant and safe altitude for a UAV's flying, which limits the sufficient utilization of a UAV's high maneuverability in space, and degrades the overall system performance. Relevant studies considering a variable flying altitude are limited.

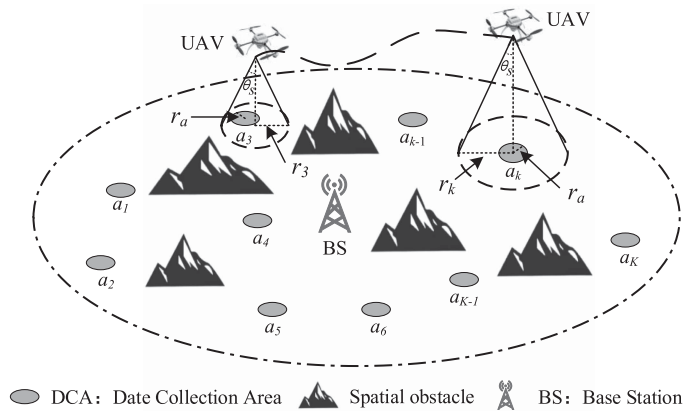

Fig. 1. System scenario.

III. System Model and Problem Formulation

This section describes the system model and formulates the 3D trajectory-planning problem under consideration.

A. System Scenario

Consider a UAV-enabled DC system in which a rotary-wing UAV is deployed to execute a DC mission in a geographical area with one BS, K DCAs and various spatial obstacles, as shown in Fig. 1. The DCAs are uniformly distributed in the area and are denoted by $\mathcal{A} = \{a_k, k = 1, \ldots, K\}$. During the process of executing the DC mission, the UAV takes off from a starting position and flies to sense and collect data over the K DCAs using onboard sensors and transmit collected data to the BS. After the DC mission is completed, the UAV will fly back to the starting position. Assume that the UAV's starting position is located at $(\mathbf{s}_0^{\mathrm{T}}, H_0)$in a 3D Cartesian coordinate system, where $\mathbf{s}_0 = [x_0, y_0]^{\mathrm{T}} \in \mathbb{R}^{2\times 1}$ denotes the starting position's projection on the ground, H_0 denotes the UAV's starting altitude and T denotes a transposition operator. The BS is located at $(0, 0, H_b)$, where H_b denotes the BS' altitude. Each DCA a_k is a small circular area with center $(\mathbf{a}_k^{\mathrm{T}}, 0)$ and radius r_a, where $\mathbf{a}_k = [x_a^k, y_a^k]^{\mathrm{T}} \in \mathbb{R}^{2\times 1}$ denotes the center's projection on the ground. The UAV's trajectory is denoted by $(\mathbf{q}(t)^{\mathrm{T}}, H(t))$,$0 \le t \le T_{total}$, where

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$\mathbf{q}(t) = [x(t), y(t)]^{\mathrm{T}} \in \mathbb{R}^{2\times 1}$ denotes the trajectory's projection on the ground, $H(t)$ denotes the UAV's flying altitude and T_{total} denotes the UAV's total mission completion time. The UAV's flying speed is limited to$v_{\max}$.

Consider that the UAV executes the DC mission in a successive-hover-fly (SHF) manner, whose effectiveness in the UAV trajectory-planning problem has been justified in [10], [18]. Thus, the process of executing the DC mission can be divided into two phases: hovering and flying.

B. Hovering Phase

The hovering phase can be further divided into two sub-phases: DC and DT.

1) DC Sub-Phase: In the DC sub-phase, the UAV needs to hover over a DCA in order to sense and collect data. We call the data collected in DCA a_k "*DATA_k*" and let $(\tilde{\mathbf{q}}_k^{\mathrm{T}}, H_q^k)$ denote the UAV's hovering position for DC in DCA a_k, where $\tilde{\mathbf{q}}_k = [x_q^k, y_q^k]^{\mathrm{T}} \in \mathbb{R}^{2\times 1}$ denotes the position's projection on the ground and H_q^k denotes the UAV's hovering altitude for DC in DCA a_k. As illustrated in Fig. 1, the UAV's sensing range on the ground is a small circular area with center position $(\tilde{\mathbf{q}}_k^{\mathrm{T}}, 0)$ and radius $r_k = H_q^k \cdot \tan\theta_s$, where θ_s denotes the maximum sensing angle of the sensor. To make sensing successful, the UAV's sensing range should cover a DCA, i.e.,

$$\|\tilde{\mathbf{q}}_k - \mathbf{a}_k\| \le r_k - r_a, k = 1, \ldots, K. \tag{1}$$

To measure if a data sensing is successful, we consider a typical probabilistic sensing model in [4], [19], in which the probability that the UAV successfully senses data in DCA a_k in one time is defined as

$$P_{suc}^1 = e^{-\tau d_q^k}, \tag{2}$$

where τ denotes a sensing factor, and d_q^k denotes the distance between the UAV hovering at $(\tilde{\mathbf{q}}_k^{\mathrm{T}}, H_q^k)$ and the center of DCA a_k. Obviously, we have

$$d_q^k = \sqrt{\|\tilde{\mathbf{q}}_k - \mathbf{a}_k\|^2 + H_q^{k2}}. \tag{3}$$

Thus, the probability that the UAV successfully senses data in DCA $a_k$in n_s times is given by

$$P_{suc}^{n_s} = 1 - \left(1 - e^{-\tau d_q^k}\right)^{n_s}. \tag{4}$$

For a successful data sensing, it is usually considered that $P_{suc}^{n_s}$should exceed a threshold ω, i.e.,

$$P_{suc}^{n_s} \ge \omega. \tag{5}$$

According to (4) and (5), the minimum time the UAV takes to successfully sense data in DCA a_k, i.e., the UAV's DC time, is given by

$$t_c^k = \breve{n}_s \times T_s = \frac{\ln(1-\omega)}{\ln[1 - e^{-\tau(\|\tilde{\mathbf{q}}_k - \mathbf{a}_k\|^2 + (H_q^k)^2)^{1/2}}]} \times T_s, \tag{6}$$

where $\breve{n}_s$ denotes the minimum number of sensing times, and T_s denotes the time the UAV takes to perform sensing per time.

2) DT Sub-Phase: In the DT sub-phase, the UAV needs to hover at a position where the SNR at the BS reaches a minimum level in order for the BS to successfully receive *DATA_k* from the UAV. For ease of exposition, let $(\tilde{\mathbf{p}}_k^{\mathrm{T}}, H_p^k)$ denote the UAV's hovering position for transmitting *DATA_k*, where $\tilde{\mathbf{p}}_k = [x_p^k, y_p^k]^{\mathrm{T}} \in \mathbb{R}^{2\times 1}$ denotes the position's projection on the ground and H_p^k denotes the UAV's hovering altitude.

Assume that the UAV hovers at $(\tilde{\mathbf{p}}_k^{\mathrm{T}}, H_p^k)$. We consider a probabilistic LoS model for the channel between the UAV and the BS [20], [21]. Let h_k denote the channel coefficient. Thus, we have

$$h_k = \sqrt{\beta_k}\tilde{h}_k, \tag{7}$$

where β_k and $\tilde{h}_k$ denote the large-scale fading and small-scale fading of the channel, respectively, and $\tilde{h}_k$ is a complex-valued random variable with $\mathbb{E}[|\tilde{h}_k|^2] = 1$. According to [20], we have

$$\beta_k = \begin{cases} \beta_0 d_p^{k^{-\alpha}}, & \text{LoSlink} \\ \kappa\beta_0 d_p^{k^{-\alpha}}, & \text{NLoSlink} \end{cases}, \tag{8}$$

where β_0 denotes the path loss at 1 meter, α denotes the path loss exponent, $\kappa(<1)$ denotes an additional attenuation factor, and d_p^k denotes the distance between the UAV and the BS. Thus, we have

$$d_p^k = \sqrt{\|\tilde{\mathbf{p}}_k\|^2 + (H_p^k - H_b)^2}. \tag{9}$$

Let P_{LoS} denote the LoS probability between the UAV and the BS. According to [20], P_{LoS} can be modeled as

$$P_{LoS}(\theta_k) = \frac{1}{1 + g_1 \exp[-g_2(\theta_k - g_1)]}, \tag{10}$$

where g_1 and g_2 denote two environmental coefficients, and θ_k denotes the elevation angle between the UAV and the BS, and$\theta_k = 180/\pi \arctan((H_p^k - H_b)/\|\tilde{\mathbf{p}}_k\|)$. Thus, the expected channel power gain between the UAV and the BS is given by

$$\begin{aligned}\mathbb{E}[|h_k|^2] &= P_{LoS}(\theta_k)\beta_0 d_p^{k^{-\alpha}} + [1 - P_{LoS}(\theta_k)]\kappa\beta_0 d_p^{k^{-\alpha}} \\ &= \hat{P}_{LoS}(\theta_k)\beta_0 d_p^{k^{-\alpha}},\end{aligned} \tag{11}$$

where

$$\widehat{P}_{LoS}(\theta_k) = P_{LoS}(\theta_k) + [1 - P_{LoS}(\theta_k)]\kappa.$$

According to [20], $\hat{P}_{LoS}(\theta_k)$ can be approximated to

$$\hat{P}_{LoS}(\theta_k) \approx \overline{P}_{LoS},$$

where $\overline{P}_{LoS}$ denotes the average LoS probability for a given UAV trajectory. Thus, (11) can approximately be expressed as

$$\mathbb{E}[|h_k|^2] \approx \overline{P}_{LoS}\beta_0 d_p^{k^{-\alpha}}. \tag{12}$$

The expected SNR at the BS is given by

$$\begin{aligned}\mathbb{E}[\gamma_k] &= \frac{p_u \mathbb{E}[|h_k|^2]}{\sigma^2\Gamma} \approx \frac{p_u \overline{P}_{LoS}\beta_0/\sigma^2\Gamma}{[\|\tilde{\mathbf{p}}_k\|^2 + (H_p^k - H_b)^2]^{\alpha/2}} \\ &= \frac{\gamma_0}{[\|\tilde{\mathbf{p}}_k\|^2 + (H_p^k - H_b)^2]^{\alpha/2}},\end{aligned} \tag{13}$$

where $\gamma_0 = p_u \overline{P}_{LoS}\beta_0/\sigma^2\Gamma$ denotes the expected SNR at 1 meter, p_u denotes the UAV's transmitting power, σ^2 denotes the noise power, and $\Gamma(>1)$ denotes an SNR gap [20]. Thus, the expected achievable rate between the UAV and the BS is approximated as

$$\begin{aligned}\mathbb{E}[R_k] &\approx B\log_2\left(1 + \frac{p_u \mathbb{E}|h_k|^2}{\sigma^2\Gamma}\right) \\ &\approx B\log_2\left\{1 + \frac{\gamma_0}{[\|\tilde{\mathbf{p}}_k\|^2 + (H_p^k - H_b)^2]^{\alpha/2}}\right\},\end{aligned} \tag{14}$$

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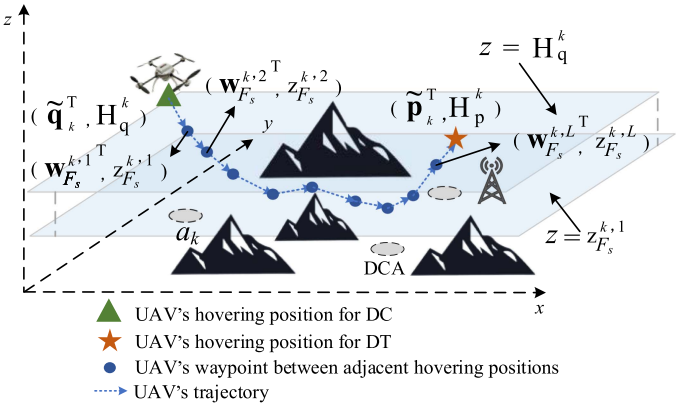

Fig. 2. The UAV's flying path with path discretization.

where B denotes the channel bandwidth allocated to the UAV.

Based on the above analyses, to achieve a successful DT, the expected SNR at the BS should meet the following condition:

$$\mathbb{E}[\gamma_k] \geq \gamma_{\min}, \tag{15}$$

where $\gamma_{\min}$ denotes an SNR threshold. The time the UAV takes to successfully transmit *DATA_k* to the BS is given by

$$t_s^k = S_k / \mathbb{E}[R_k], \tag{16}$$

where S_k denotes the size of *DATA_k*.

C. Flying Phase

In the flying phase, the UAV successively flies to visit its hovering positions for DC $\{(\tilde{\mathbf{q}}_k^{\mathrm{T}}, H_q^k), k = 1, \ldots, K\}$ and hovering positions for DT $\{(\tilde{\mathbf{p}}_k^{\mathrm{T}}, H_p^k), k = 1, \ldots, K\}$ in a certain sequence. Let $\pi(k) \in \{1, \ldots, K\}$ denote the index of the kth visited DCA. Thus, the UAV's visiting sequence is given by $[\pi(1), \pi(2), \ldots, \pi(K)] \in \Psi$, where Ψ denotes a set of all possible permutations. Since there are K DCAs, Ψ is actually a set of $K!$ random permutations of integers from 1 to K, where $K!$ denotes the factorial of K.

Unlike [14], in which a 2D scenario is considered, we consider a 3D scenario where the UAV does not fly along a straight line between two adjacent hovering positions at a constant and safe altitude. Since there may exist obstacles between two adjacent hovering positions, the UAV needs to fly around or over these obstacles to avoid collisions, resulting in the trajectory planning between two adjacent hovering positions more complicated. To make the trajectory planning tractable, a path discretization method [5], [20] is introduced to discretize the UAV's flying path between two adjacent hovering positions into $L+1$ path segments with L waypoints, as illustrated in Fig. 2. The Lwaypoints are determined such that by flying along the waypoints of the path, the UAV can circumvent or fly over these obstacles. Obviously, if L is large enough, the path length between two adjacent hovering positions would be much larger than the length of a path segment. Thus, the UAV can be considered flying straightly at a constant speed on each path segment. Since the durations the UAV flies on different path segments are different, the UAV's flying path between two adjacent hovering positions can be represented by the UAV's waypoints and flying durations on all segments of the flying path.

There are three situations for the UAV's trajectory between two adjacent hovering positions:

S-1: from position $(\tilde{\mathbf{p}}_{\pi(\pi'(k)-1)}, H_p^{\pi(\pi'(k)-1)})$ to position $(\tilde{\mathbf{q}}_k^{\mathrm{T}}, H_q^k)$. In this situation, the UAV flies from its hovering position for transmitting data collected in DCA $a_{\pi(\pi'(k)-1)}$ to its hovering position for DC in DCA a_k, where $\pi'(k)$ is the inverse function of $\pi(k)$ and denotes the UAV's visiting order for DCA a_k, and $a_{\pi(\pi'(k)-1)}$ is the last visited DCA before DCA a_k is visited by the UAV;

S-2: from position $(\tilde{\mathbf{q}}_k^{\mathrm{T}}, H_q^k)$ to position $(\tilde{\mathbf{p}}_k^{\mathrm{T}}, H_p^k)$. In this situation, the UAV flies from its hovering position for DC in DCA a_k to its hovering position for transmitting *DATA_k*;

S-3: from position $(\tilde{\mathbf{p}}_{\pi(K)}^{\mathrm{T}}, H_p^{\pi(K)})$ to position $(\mathbf{s}_0{}^{\mathrm{T}}, H_0)$. In this situation, the UAV flies from its hovering position for transmitting data collected in the last DCA $a_{\pi(K)}$ to its starting position.

Note that in S-1, when $\pi'(k) = 1$, the UAV flies from its starting position to visit DCA a_k first. For ease of exposition, we define $(\tilde{\mathbf{p}}_{\pi(0)}^{\mathrm{T}}, H_p^{\pi(0)}) = (\mathbf{s}_0{}^{\mathrm{T}}, H_0)$, i.e., $(\tilde{\mathbf{p}}_{\pi(0)}^{\mathrm{T}}, H_p^{\pi(0)})$ also denotes the UAV's starting position.

Corresponding to the three situations, we introduce three types of waypoints to discretize the UAV's path between two adjacent hovering positions, respectively, i.e., the waypoints for collecting *DATA_k*, waypoints for transmitting *DATA_k*, and waypoints for flying back to the starting position. Let$(\mathbf{w}_{F_c}^{k,n\,\mathrm{T}}, z_{F_c}^{k,n})$, $(\mathbf{w}_{F_s}^{k,n\,\mathrm{T}}, z_{F_s}^{k,n})$ and $(\mathbf{w}_{F_0}^{n\,\mathrm{T}}, z_{F_0}^{n})$,$k = 1, \ldots, K$,$n = 1, \ldots, L$, denote the 3D coordinates of the three types of waypoints, respectively, where $\mathbf{w}_{F_c}^{k,n} = [x_{F_c}^{k,n}, y_{F_c}^{k,n}]^{\mathrm{T}} \in \mathbb{R}^{2\times 1}$, $\mathbf{w}_{F_s}^{k,n} = [x_{F_s}^{k,n}, y_{F_s}^{k,n}]^{\mathrm{T}} \in \mathbb{R}^{2\times 1}$ and $\mathbf{w}_{F_0}^{n} = [x_{F_0}^{n}, y_{F_0}^{n}]^{\mathrm{T}} \in \mathbb{R}^{2\times 1}$ denote their horizontal coordinates, respectively;$z_{F_c}^{k,n}$, $z_{F_s}^{k,n}$ and $z_{F_0}^{n}$ denote their altitudes, respectively. The UAV's durations on the nth path segments in the three situations are $t_{F_c}^{k,n}$, $t_{F_s}^{k,n}$ and $t_{F_0}^{n}$, respectively. Since there are $L+1$ path segments, we have $n = 1, \ldots, L+1$. Let$v_{F_c}^{k,n}$, $v_{F_s}^{k,n}$ and $v_{F_0}^{n}$ denote the UAV's instantaneous flying speeds on the nth path segments in the three situations, respectively. Obviously, we have

$$v_{F_c}^{k,n} = \frac{\sqrt{\left\|\mathbf{w}_{F_c}^{k,n-1} - \mathbf{w}_{F_c}^{k,n}\right\|^2 + \left(z_{F_c}^{k,n-1} - z_{F_c}^{k,n}\right)^2}}{t_{F_c}^{k,n}} = \frac{\Delta_{F_c}^{k,n}}{t_{F_c}^{k,n}}, \tag{17}$$

$$v_{F_s}^{k,n} = \frac{\sqrt{\left\|\mathbf{w}_{F_s}^{k,n-1} - \mathbf{w}_{F_s}^{k,n}\right\|^2 + \left(z_{F_s}^{k,n-1} - z_{F_s}^{k,n}\right)^2}}{t_{F_s}^{k,n}} = \frac{\Delta_{F_s}^{k,n}}{t_{F_s}^{k,n}}, \tag{18}$$

$$v_{F_0}^{n} = \frac{\sqrt{\left\|\mathbf{w}_{F_0}^{n-1} - \mathbf{w}_{F_0}^{n}\right\|^2 + \left(z_{F_0}^{n-1} - z_{F_0}^{n}\right)^2}}{t_{F_0}^{n}} = \frac{\Delta_{F_0}^{n}}{t_{F_0}^{n}}, \tag{19}$$

where $(\mathbf{w}_{F_c}^{k,0}, z_{F_c}^{k,0}) = (\tilde{\mathbf{p}}_{\pi(\pi'(k)-1)}, H_p^{\pi(\pi'(k)-1)})$, $(\mathbf{w}_{F_c}^{k,L+1}, z_{F_c}^{k,L+1}) = (\tilde{\mathbf{q}}_k, H_q^k)$, $(\mathbf{w}_{F_s}^{k,0}, z_{F_s}^{k,0}) = (\tilde{\mathbf{q}}_k, H_q^k)$, $(\mathbf{w}_{F_s}^{k,L+1}, z_{F_s}^{k,L+1}) = (\tilde{\mathbf{p}}_k, H_p^k)$,$(\mathbf{w}_{F_0}^{0}, z_{F_0}^{0}) = (\tilde{\mathbf{p}}_{\pi(K)}, H_p^{\pi(K)})$ $(\mathbf{w}_{F_0}^{L+1}, z_{F_0}^{L+1}) = (\mathbf{s}_0, H_0)$, $k = 1, \ldots, K$ and $n = 1, \ldots, L+1$, and $\Delta_{F_c}^{k,n}$, $\Delta_{F_s}^{k,n}$ as well as $\Delta_{F_0}^{n}$ denote the length of each path segment in S-1, S-2, and S-3, respectively. Since the UAV's

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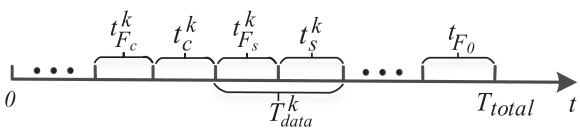

Fig. 3. The UAV's mission execution time sequence.

maximum flying speed is $v_{\max}$, we have

$$\Delta_{F_c}^{k,n} \leq v_{\max} t_{F_c}^{k,n}, \tag{20}$$

$$\Delta_{F_s}^{k,n} \leq v_{\max} t_{F_s}^{k,n}, \tag{21}$$

$$\Delta_{F_0}^{n} \leq v_{\max} t_{F_0}^{n}, \tag{22}$$

where $n = 1, \ldots, L+1$ and $k = 1, \ldots, K$.

D. Time Sequence and Time Constraint

Based on the analyses in Section III-B and III-C, the UAV's total mission completion time can be expressed as

$$T_{total} = \sum_{k=1}^{K} (t_{F_c}^k + t_c^k + t_{F_s}^k + t_s^k) + t_{F_0}, \tag{23}$$

and $t_{F_c}^k = \sum_{n=1}^{L+1} t_{F_c}^{k,n}$, $t_{F_s}^k = \sum_{n=1}^{L+1} t_{F_s}^{k,n}$, $t_{F_0} = \sum_{n=1}^{L+1} t_{F_0}^{n}$, where $t_{F_c}^k$,t_c^k,$t_{F_s}^k$,t_s^k,t_{F_0},$k = 1, \ldots, K$, denote the UAV's flying time for collecting *DATA_k*, collecting time for *DATA_k*, flying time for transmitting *DATA_k*, transmitting time for *DATA_k*, and flying time for flying back to the starting position, respectively.

Fig. 3 illustrates the UAV's mission execution time sequence. Let T_{data}^k denote the age of *DATA_k*, i.e., the time interval from the instant the UAV collects *DATA_k* to the instant the BS successfully receives *DATA_k*. According to Fig. 3, we have

$$T_{data}^k = t_s^k + t_{F_s}^k. \tag{24}$$

To ensure that *DATA_k* is still effective after being received by the BS, T_{data}^k should meet the following condition:

$$T_{data}^k \leq T_{limit}^k, k = 1, \ldots, K, \tag{25}$$

where T_{limit}^k denotes the time constraint of *DATA_k*.

E. Energy Constraint Model

The UAV's energy consumption includes that for DC, that for DT, and that for propulsion. Let P_c and P_s denote the UAV's power consumption for DC and that for DT, respectively. Thus, the energy consumption models for DC and DT are respectively given by

$$E_c = \sum_{k=1}^{K} (P_c t_c^k), \tag{26}$$

$$E_s = \sum_{k=1}^{K} (P_s t_s^k). \tag{27}$$

According to [22], for a 3D UAV trajectory $(\mathbf{q}(t)^{\mathrm{T}}, H(t))$, the UAV's energy consumption for propulsion can be approximated to

$$E_p \approx \int_0^{T_{total}} P_p(v(t))dt + \frac{1}{2} M_u (v(T_{total})^2 - v(0)^2) + M_u g(H(T_{total}) - H(0)), \tag{28}$$

where $v(t)$ denotes the UAV's instantaneous speed, M_u denotes the UAV's weight, g denotes the gravitational acceleration and $P_p(v(t))$ denotes the UAV's power consumption for propulsion. For a rotary-wing UAV, $P_p(v(t))$ is given by

$$P_p(v(t)) = c_1(1 + c_2 v(t)^2) + c_3\left(\sqrt{1 + \frac{v(t)^4}{(c_4)^2}} - \frac{v(t)^2}{c_4}\right)^{1/2} + c_5 v(t)^3, \tag{29}$$

where $c_i, i = 1, \ldots, 5$ denotes a set of parameters which depend on the UAV's characteristics and environmental conditions, as described in [22].

Since the UAV flies back to the starting position after completing the overall mission, and the UAV's starting speed and ending speed are equal to 0, we have

$$v(T_{total}) = v(0) = 0,$$

$$H(T_{total}) = H(0) = H_0.$$

By substituting these two equations into (28), (28) can be further expressed as

$$E_p \approx \int_0^{T_{total}} P_p(v(t))dt. \tag{30}$$

Moreover, since the UAV executes the overall mission in a successive-hover-fly manner, the UAV's energy consumption for propulsion can also be expressed as

$$E_p = E_h + E_f, \tag{31}$$

where E_h and E_f denote the UAV's energy consumption for hovering and that for flying, respectively.

By substituting $v(t) = 0$ into (29), we can obtain the UAV's power consumption for hovering, i.e.,

$$P_h = c_1 + c_3.$$

Thus, the UAV's energy consumption for hovering is given by

$$E_h = \sum_{k=1}^{K} P_h(t_c^k + t_s^k). \tag{32}$$

According to the analyses in Section III-C and Fig. 3, the UAV's energy consumption for flying is given by

$$E_f = \sum_{k=1}^{K} \sum_{n=1}^{L+1} [P_p(v_{F_c}^{k,n}) t_{F_c}^{k,n} + P_p(v_{F_s}^{k,n}) t_{F_s}^{k,n}] + \sum_{n=1}^{L+1} [P_p(v_{F_0}^{n}) t_{F_0}^{n}]. \tag{33}$$

Therefore, the UAV's total energy consumption is

$$E_{total} = E_p + E_c + E_s, \tag{34}$$

and the UAV's energy constraint model is

$$E_{total} \leq E_{limit}, \tag{35}$$

where E_{limit} denotes the UAV's maximum available energy.

F. Obstacle Model

We consider a typical obstacle model [5], i.e.,

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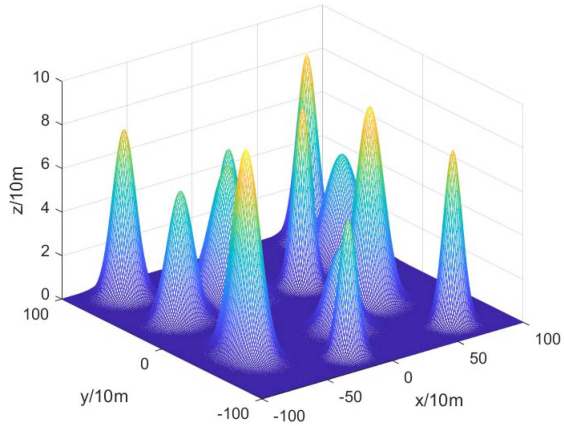

Fig. 4. A map of the obstacle model for a geographical area.

$$z_{obs}(x,y) = \sum_{i=1}^{G}\left\{h_{obs}^{i} \times \exp\left[-\left(\frac{x-x_{obs}^{i}}{a_x^i}\right)^2 - \left(\frac{y-y_{obs}^{i}}{a_y^i}\right)^2\right]\right\}, \tag{36}$$

where $i = 0, \ldots, G$, (x, y) denotes the horizontal coordinate of a particular position, $z_{obs}(x, y)$ denotes the altitude of the obstacle at the position (x, y), G denotes the number of obstacles in the area, h_{obs}^i denotes the altitude of the ith obstacle, (x_{obs}^i, y_{obs}^i) denotes the horizontal coordinate of the ith obstacle's center, a_x^i and a_y^i denote the ith obstacle's attenuation on axis x and axis y, respectively. Fig. 4 illustrates a map of the obstacle model for the geographical area.

In the UAV flying process, the UAV should avoid collisions with obstacles to ensure its flying safety. Thus, the UAV's flying altitude at each hovering position or waypoint should be larger than the altitude of the obstacle at that position or waypoint during the overall mission, i.e.,

$$z_{obs}(x_q^k, y_q^k) < H_q^k, \tag{37}$$

$$z_{obs}(x_p^k, y_p^k) < H_p^k, \tag{38}$$

$$z_{obs}(x_{F_c}^{k,n}, y_{F_c}^{k,n}) < z_{F_c}^{k,n}, \tag{39}$$

$$z_{obs}(x_{F_s}^{k,n}, y_{F_s}^{k,n}) < z_{F_s}^{k,n}, \tag{40}$$

$$z_{obs}(x_{F_0}^{n}, y_{F_0}^{n}) < z_{F_0}^{n}, \tag{41}$$

where $n = 1, \ldots, L+1$ and $k = 1, \ldots, K$.

G. Problem Formulation

We consider the UAV trajectory-planning problem in a UAV-enabled DC system shown in Fig. 1. Since the UAV executes the overall mission in a successive-hover-fly manner, the UAV's trajectory can be represented by $\boldsymbol{\Omega} = \{\boldsymbol{\mathcal{Q}}, \boldsymbol{\pi}, \boldsymbol{\mathcal{W}}, \boldsymbol{\mathcal{T}}\}$, where $\boldsymbol{\pi}$ denotes the UAV's visiting sequence, $\boldsymbol{\mathcal{Q}}$ denotes a set of the UAV's hovering positions, $\boldsymbol{\mathcal{W}}$ denotes a set of waypoints on each flying path between different hovering positions, and $\boldsymbol{\mathcal{T}}$ denotes a set of flying durations on all segments of each flying path between different hovering positions. Obviously, we have

$$\boldsymbol{\pi} = [\pi(1), \ldots, \pi(K)],$$

$$\boldsymbol{\mathcal{Q}} = \{\tilde{\mathbf{q}}_k, \tilde{\mathbf{p}}_k, H_q^k, H_p^k, k = 1, \ldots, K\},$$

$$\boldsymbol{\mathcal{W}} = \{\mathbf{w}_{F_c}^{k,n}, \mathbf{w}_{F_s}^{k,n}, \mathbf{w}_{F_0}^{n}, z_{F_c}^{k,n}, z_{F_s}^{k,n}, z_{F_0}^{n}, k = 1, \ldots, K, n = 1, \ldots, L\},$$

$$\boldsymbol{\mathcal{T}} = \{t_{F_s}^{k,n}, t_{F_s}^{k,n}, t_{F_0}^{n}, k = 1, \ldots, K, n = 1, \ldots, L+1\}.$$

Let $\boldsymbol{\Omega}^*$ denote the UAV's optimal trajectory. The UAV trajectory-planning problem under consideration is to find $\boldsymbol{\Omega}^*$ that minimizes the UAV's total mission completion time with the time constraint of collected data, and the avoidance of obstacles, and the energy constraint of the UAV, i.e.,

P0:

Objective:

$$\min_{\boldsymbol{\Omega}} T_{total} \tag{42a}$$

subject to:

$$[\pi(1), \pi(2), \ldots, \pi(K)] \in \Psi, \tag{42b}$$

$$(1), (15), (20)\text{–}(22), (25), (35), (37)\text{–}(41),$$

where (42b) guarantees that the UAV's visiting sequence should be a random permutation of integers from 1 to K, (1) guarantees that the UAV's sensing range should cover DCA a_k, (15) guarantees that the SNR at the BS should exceed a threshold, (20)–(22) guarantee that the UAV's flying speed should not exceed its maximum speed, (25) indicates the time constraint of *DATA_k*, (35) indicates the UAV's energy constraint, (37)–(41) guarantee that the UAV's flying altitude at each hovering position or waypoint should be higher than the altitude of the obstacle at that position or waypoint during the overall mission.

Since the objective function (42a) and constraints (25), (35), (37)–(41) and (42b) of P0 are non-convex, P0 is a non-convex problem. Thus, it is hard to find an optimal solution to the problem. Next, we will present a 3D trajectory optimization algorithm to find a sub-optimal solution to the problem.

IV. 3D Trajectory Optimization Algorithm

In this section, we first decompose the formulated problem into a visiting sequence optimization sub-problem, and a hovering position and flying path optimization sub-problem, and then propose two algorithms to solve the sub-problems, respectively, and a 3D trajectory optimization algorithm to solve the main problem.

A. UAV Visiting Sequence Optimization

Let $\boldsymbol{\pi}^*$ denote the UAV's optimal visiting sequence. Given the UAV's hovering positions and flying paths between the hovering positions, the visiting sequence optimization problem is to find $\boldsymbol{\pi}^*$ that minimizes the UAV's total mission completion time, i.e.,

P1:

Objective:

$$\min_{\boldsymbol{\pi}} T_{total} \tag{43a}$$

subject to:

$$(42b), (25), (35).$$

According to (43), P1 is actually an asymmetric traveling salesman problem, which can be solved by using an existing algorithm, such as a greedy algorithm, heuristic algorithm, or dynamic programming. Similar to [14], we use a GA-based optimization algorithm to solve P1, which is described in Algorithm 1.

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Algorithm 1: GA-Based Optimization Algorithm.

Input: a set of UAV's hovering positions: $\boldsymbol{\mathcal{Q}} = \{\tilde{\mathbf{q}}_k, \tilde{\mathbf{p}}_k, H_q^k, H_p^k, k = 1, \ldots, K\}$,
a set of a waypoints on each flying path: $\boldsymbol{\mathcal{W}} = \{\mathbf{w}_{F_c}^{k,n}, \mathbf{w}_{F_s}^{k,n}, \mathbf{w}_{F_0}^{n}, z_{F_c}^{k,n}, z_{F_s}^{k,n}, z_{F_0}^{n}, k = 1, \ldots, K, n = 1, \ldots, L\}$,
a set of flying durations on all segments: $T = \{t_{F_s}^{k,n}, t_{F_s}^{k,n}, t_{F_0}^{n}, k = 1, \ldots, K, n = 1, \ldots, L+1\}$
a set of AoI limitations: $\{T_{limit}^k, k = 1, \ldots, K\}$,
GA-related parameters: the initial population size Z, the chromosome selection probability P_{select}, the crossover probability of two random chromosomes P_{cross}, the mutation probability of a gene P_m and the maximum predetermined number of iterations Y;
the maximum on-board energy of the UAV: E_{limit}.

1: Generate an initial population with Z chromosomes randomly with each chromosome satisfying the constraints of problem P1;
2: Set the iteration variable: $l = 0$;
3: **Repeat**
4: Calculate the fitness values of Z chromosomes according to (43a);
5: Select part of the best chromosomes to directly pass down to the next generation with probability P_{select} according to the fitness values of Z chromosomes;
6: Perform crossover operations on two random chromosomes with probability P_{cross} and produce new chromosomes;
7: Perform mutation operations on Z chromosomes with probability P_m and produce new chromosomes;
8: Check whether each of the new chromosomes satisfies the constraints of problem P1. If it does, retain it; otherwise, discard it;
9: Generate a new population with new Z chromosomes and calculate the fitness value of each chromosome according to (43a);
10: Perform $l = l + 1$;
11: **Until** $l \geq Y$
12: Obtain the final population with Z chromosomes and the fitness value of each chromosome;
13: Choose the chromosome with the minimum fitness value as the optimal UAV's visiting sequence;

Output: the optimal UAV's visiting sequence: $\boldsymbol{\pi}^* = [\pi^*(1), \ldots, \pi^*(K)]$.

B. UAV Hovering Position and Flying Path Optimization

Let $\boldsymbol{\mathcal{Q}}^*$, $\boldsymbol{\mathcal{W}}^*$, and $\boldsymbol{\mathcal{T}}^*$ denote the optimal $\boldsymbol{\mathcal{Q}}$, $\boldsymbol{\mathcal{W}}$, and $\boldsymbol{\mathcal{T}}$, respectively. Given the UAV's visiting sequence, the UAV hovering position and flying path optimization problem is to find $\{\boldsymbol{\mathcal{Q}}^*, \boldsymbol{\mathcal{W}}^*, \boldsymbol{\mathcal{T}}^*\}$ that minimizes the UAV's total mission completion time, i.e.,

P2

Objective:

$$\min_{\boldsymbol{\mathcal{Q}}, \boldsymbol{\mathcal{W}}, \boldsymbol{\mathcal{T}}} T_{total} \tag{44a}$$

subject to:

$$(1), (15), (20)\text{–}(22), (25), (35), (37)\text{–}(41).$$

Since constraints (25), (35), (37)–(41) are all non-convex, P2 is a typical non-convex problem. Thus, it is hard to find an optimal solution to the problem.

To solve the problem, we first consider the non-convex objective function (44a). According to (6), (16) and (23), we have

$$\begin{aligned} T_{total} = &\sum_{k=1}^{K}\sum_{n=1}^{L+1} t_{F_c}^{k,n} + \sum_{k=1}^{K}\sum_{n=1}^{L+1} t_{F_s}^{k,n} + \sum_{n=1}^{L+1} t_{F_0}^{n} \\ &+ \sum_{k=1}^{K}\left(\frac{\ln(1-\omega)}{\ln[1 - e^{-\tau(\|\tilde{\mathbf{q}}_k - \mathbf{a}_k\|^2 + H_q^{k\,2})^{1/2}}]} \times T_s\right) \\ &+ \sum_{k=1}^{K}\left(\frac{S_k/B}{\log_2\{1 + \gamma_0/[\|\tilde{\mathbf{p}}_k\|^2 + (H_p^k - H_b)^2]^{\alpha/2}\}}\right). \end{aligned} \tag{45}$$

In (45), the fourth and fifth terms are non-convex. To tackle these two terms, we introduce two sets of slack variables:$\{\varphi_k\}$,$\{\eta_k\}$,$k = 1, \ldots, K$, where

$$\varphi_k = \ln[\frac{1}{1 - e^{-\tau(\|\tilde{\mathbf{q}}_k - \mathbf{a}_k\|^2 + (H_q^k)^2)^{1/2}}}], \tag{46}$$

$$\eta_k = \log_2\{1 + \frac{\gamma_0}{[\|\tilde{\mathbf{p}}_k\|^2 + (H_p^k - H_b)^2)]^{\alpha/2}}\}. \tag{47}$$

Thus, the fourth and fifth terms in (45) can be replaced by convex terms $\sum_{k=1}^{K}[T_s \ln(\frac{1}{1-\omega})/\varphi_k]$ and $\sum_{k=1}^{K}[S_k/(B\eta_k)]$, respectively, with additional constraints (46) and (47). Accordingly, constraint (25) can be replaced by

$$\frac{S_k}{B\eta_k} + \sum_{n=1}^{L+1} t_{F_s}^{k,n} \leq T_{limit}^k, \tag{48}$$

which is a convex constraint.

Next, we consider constraint (35). According to the analyses in Section III-E, constraint (35) can be extended to constraint (49) shown at the bottom of the next page, which contains the following three non-convex terms:

$$\sqrt{(t_{F_c}^{k,n})^4 + \frac{(\Delta_{F_c}^{k,n})^4}{(c_4)^2}} - \frac{(\Delta_{F_c}^{k,n})^2}{c_4},$$

$$\sqrt{(t_{F_s}^{k,n})^4 + \frac{(\Delta_{F_s}^{k,n})^4}{(c_4)^2}} - \frac{(\Delta_{F_s}^{k,n})^2}{c_4},$$

$$\sqrt{(t_{F_0}^{n})^4 + \frac{(\Delta_{F_0}^{n})^4}{(c_4)^2}} - \frac{(\Delta_{F_0}^{n})^2}{c_4}.$$

To tackle these three terms, we introduce three sets of slack variables:$\{\mu_{F_c}^{k,n}\}$,$\{\mu_{F_s}^{k,n}\}$,$\{\mu_{F_0}^{n}\}$,$k = 1, .., K$,$n = 1, \ldots, L+1$,

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where

$$(t_{F_c}^{k,n})^4/(\mu_{F_c}^{k,n})^2 = (\mu_{F_c}^{k,n})^2 + 2(\Delta_{F_c}^{k,n})^2/c_4, \tag{50}$$

$$(t_{F_s}^{k,n})^4/(\mu_{F_s}^{k,n})^2 = (\mu_{F_s}^{k,n})^2 + 2(\Delta_{F_s}^{k,n})^2/c_4, \tag{51}$$

$$(t_{F_0}^{n})^4/(\mu_{F_0}^{n})^2 = (\mu_{F_0}^{n})^2 + 2(\Delta_{F_0}^{n})^2/c_4. \tag{52}$$

Thus, the three terms can be replaced by convex terms $\mu_{F_c}^{k,n}$, $\mu_{F_s}^{k,n}$ and $\mu_{F_0}^{n}$, respectively, with additional constraints (50), (51) and (52). Finally, we consider constraints (37)–(41). Since constraints (37)–(41) are similar, we only discuss constraint (37). According to the analyses in Section III-F, we have

$$\sum_{i=1}^{G}\left\{h_{obs}^{i}\times\exp\left[-\left(\frac{x_q^k-x_{obs}^i}{a_x^i}\right)^2-\left(\frac{y_q^k-y_{obs}^i}{a_y^i}\right)^2\right]\right\}<H_q^k. \tag{53}$$

Note that in (53), the exponential term $\exp[-(\frac{x_q^k-x_{obs}^i}{a_x^i})^2-(\frac{y_q^k-y_{obs}^i}{a_y^i})^2]$ is non-convex. To tackle this term, we introduce a set of slack variable:$\{\psi_q^{i,k}\}$, $i=1,..,G$,$k=1,\ldots,K$, where

$$\psi_q^{i,k}=\left(\frac{x_q^k-x_{obs}^i}{a_x^i}\right)^2+\left(\frac{y_q^k-y_{obs}^i}{a_y^i}\right)^2. \tag{54}$$

Thus, the exponential term can be replaced by convex term $e^{-\psi_q^{i,k}}$ with additional constraint (54). Accordingly, for constraints (38)–(41), we introduce another four sets of slack variables:$\{\psi_p^{i,k}\}$,$\{\psi_{F_c}^{i,k,n}\}$,$\{\psi_{F_s}^{i,k,n}\}$,$\{\psi_{F_0}^{i,n}\}$,$i=1,..,G$,$k=1,\ldots,K$,$n=1,..,L+1$. The non-convex exponential terms in constraints (38)–(41) can be replaced by convex terms $e^{-\psi_p^{i,k}}$,$e^{-\psi_{F_c}^{i,k,n}}$,$e^{-\psi_{F_s}^{i,k,n}}$ and $e^{-\psi_{F_0}^{i,n}}$, respectively, with additional constraints. Since $\{\psi_p^{i,k}\}$,$\{\psi_{F_c}^{i,k,n}\}$,$\{\psi_{F_s}^{i,k,n}\}$, and $\{\psi_{F_0}^{i,n}\}$ all have the similar form to that of$\{\psi_q^{i,k}\}$, for ease of exposition, we will not unnecessarily give the formulas of $\{\psi_p^{i,k}\}$,$\{\psi_{F_c}^{i,k,n}\}$,$\{\psi_{F_s}^{i,k,n}\}$, and $\{\psi_{F_0}^{i,n}\}$here.

Based on the above analyses, let $\mathbf{L}$ denote a set of all the introduced slack variables, i.e.,

$$\mathbf{L}=\{\varphi_k,\eta_k,\mu_{F_c}^{k,n},\mu_{F_s}^{k,n},\mu_{F_0}^{n},\psi_q^{i,k},\psi_p^{i,k},\psi_{F_c}^{i,k,n},\psi_{F_s}^{i,k,n},\psi_{F_0}^{i,n},$$
$$i=1,..,G,k=1,\ldots,K,n=1,..,L+1\}.$$

Then, P2 can be equivalently formulated as P2.1, i.e.,

P2.1

Objective:

$$\min_{\mathcal{Q},W,\mathcal{T},\mathbf{L}}\left\{\sum_{k=1}^{K}\left(\begin{array}{l}\sum_{n=1}^{L+1}t_{F_c}^{k,n}+\sum_{n=1}^{L+1}t_{F_s}^{k,n}+\frac{S_k}{B\eta_k}\\+T_s\ln(\frac{1}{1-\omega})/\varphi_k\end{array}\right)+\sum_{n=1}^{L+1}t_{F_0}^{n}\right\}, \tag{55a}$$

subject to:

$$\frac{S_k}{B\eta_k}+\sum_{n=1}^{L+1}t_{F_s}^{k,n}\le T_{limit}^k, \tag{55b}$$

$$\begin{aligned}&\sum_{k=1}^{K}\sum_{n=1}^{L+1}c_1\left[t_{F_c}^{k,n}+t_{F_s}^{k,n}+c_2\left(\frac{(\Delta_{F_c}^{k,n})^2}{t_{F_c}^{k,n}}+\frac{(\Delta_{F_s}^{k,n})^2}{t_{F_s}^{k,n}}\right)\right]\\&+\sum_{k=1}^{K}\sum_{n=1}^{L+1}c_3(\mu_{F_c}^{k,n}+\mu_{F_s}^{k,n})+\sum_{k=1}^{K}\sum_{n=1}^{L+1}c_5\left(\frac{(\Delta_{F_c}^{k,n})^3}{(t_{F_c}^{k,n})^2}+\frac{(\Delta_{F_s}^{k,n})^3}{(t_{F_s}^{k,n})^2}\right)\\&+\sum_{n=1}^{L+1}c_1(t_{F_0}^{n}+c_2\frac{(\Delta_{F_0}^{n})^2}{t_{F_0}^{n}})+\sum_{n=1}^{L+1}c_3\mu_{F_0}^{n}+\sum_{n=1}^{L+1}c_5\frac{(\Delta_{F_0}^{n})^3}{(t_{F_0}^{n})^2}\\&+\sum_{k=1}^{K}P_h\left(T_s\ln\left(\frac{1}{1-\omega}\right)/\varphi_k+\frac{S_k/B}{\eta_k}\right)\\&+\sum_{k=1}^{K}P_cT_s\ln\left(\frac{1}{1-\omega}\right)/\varphi_k+\sum_{k=1}^{K}P_s\frac{S_k/B}{\eta_k}\le E_{limit},\end{aligned} \tag{55c}$$

$$\sum_{i=1}^{G}(h_{obs}^{i}\times e^{-\psi_q^{i,k}})<H_q^k, \tag{55d}$$

$$\sum_{i=1}^{G}(h_{obs}^{i}\times e^{-\psi_p^{i,k}})<H_p^k, \tag{55e}$$

$$\sum_{i=1}^{G}(h_{obs}^{i}\times e^{-\psi_{F_c}^{i,k,n}})<z_{F_c}^{k,n}, \tag{55f}$$

$$\sum_{i=1}^{G}(h_{obs}^{i}\times e^{-\psi_{F_s}^{i,k,n}})<z_{F_s}^{k,n}, \tag{55g}$$

$$\begin{aligned}&\sum_{k=1}^{K}\sum_{n=1}^{L+1}c_1\left(t_{F_c}^{k,n}+t_{F_s}^{k,n}+c_2\left[\frac{(\Delta_{F_c}^{k,n})^2}{t_{F_c}^{k,n}}+\frac{(\Delta_{F_s}^{k,n})^2}{t_{F_s}^{k,n}}\right]\right)+\sum_{k=1}^{K}\sum_{n=1}^{L+1}c_3\left(\left[\sqrt{(t_{F_c}^{k,n})^4+\frac{(\Delta_{F_c}^{k,n})^4}{{c_4}^2}}-\frac{(\Delta_{F_c}^{k,n})^2}{c_4}\right]^{1/2}\right.\\&\left.+\left[\sqrt{(t_{F_s}^{k,n})^4+\frac{(\Delta_{F_s}^{k,n})^4}{{c_4}^2}}-\frac{(\Delta_{F_s}^{k,n})^2}{c_4}\right]^{1/2}\right)+\sum_{k=1}^{K}\sum_{n=1}^{L+1}c_5\left[\frac{(\Delta_{F_c}^{k,n})^3}{(t_{F_c}^{k,n})^2}+\frac{(\Delta_{F_s}^{k,n})^3}{(t_{F_s}^{k,n})^2}\right]+\sum_{n=1}^{L+1}c_1\left[t_{F_0}^{n}+c_2\frac{(\Delta_{F_0}^{n})^2}{t_{F_0}^{n}}\right]\\&+\sum_{n=1}^{L+1}c_3\left[\sqrt{(t_{F_0}^{n})^4+\frac{(\Delta_{F_0}^{n})^4}{{c_4}^2}}-\frac{(\Delta_{F_0}^{n})^2}{c_4}\right]^{1/2}+\sum_{n=1}^{L+1}c_5\frac{(\Delta_{F_0}^{n})^3}{(t_{F_0}^{n})^2}+\sum_{k=1}^{K}P_h\left[T_s\ln\left(\frac{1}{1-\omega}\right)\Big/\varphi_k+\frac{S_k/B}{\eta_k}\right]\\&+\sum_{k=1}^{K}P_cT_s\ln\left(\frac{1}{1-\omega}\right)\Big/\varphi_k+\sum_{k=1}^{K}P_s\frac{S_k/B}{\eta_k}\le E_{limit}.\end{aligned} \tag{49}$$

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$$\sum_{i=1}^{G}(h_{obs}^{i}\times e^{-\psi_{F_0}^{i,n}})<z_{F_0}^{n}, \tag{55h}$$

$$\varphi_k \le \ln\left[\frac{1}{1-e^{-\tau(\|\tilde{\mathbf{q}}_k-\mathbf{a}_k\|^2+(H_q^k)^2)^{1/2}}}\right], \tag{55i}$$

$$\eta_k \le \log_2\left\{1+\frac{\gamma_0}{[\|\tilde{\mathbf{p}}_k\|^2+(H_p^k-H_b)^2]^{\alpha/2}}\right\}, \tag{55j}$$

$$(t_{F_c}^{k,n})^4/(\mu_{F_c}^{k,n})^2 \le (\mu_{F_c}^{k,n})^2+2(\Delta_{F_c}^{k,n})^2/c_4, \tag{55k}$$

$$(t_{F_s}^{k,n})^4/(\mu_{F_s}^{k,n})^2 \le (\mu_{F_s}^{k,n})^2+2(\Delta_{F_s}^{k,n})^2/c_4, \tag{55l}$$

$$(t_{F_0}^{n})^4/(\mu_{F_0}^{n})^2 \le (\mu_{F_0}^{n})^2+2(\Delta_{F_0}^{n})^2/c_4, \tag{55m}$$

$$\psi_{q,i}^{k} \le \left(\frac{x_q^k-x_{obs}^i}{a_x^i}\right)^2+\left(\frac{y_q^k-y_{obs}^i}{a_y^i}\right)^2, \tag{55n}$$

$$\psi_{p,i}^{k} \le \left(\frac{x_p^k-x_{obs}^i}{a_x^i}\right)^2+\left(\frac{y_p^k-y_{obs}^i}{a_y^i}\right)^2, \tag{55o}$$

$$\psi_{F_c,i}^{k,n} \le \left(\frac{x_{F_c}^{k,n}-x_{obs}^i}{a_x^i}\right)^2+\left(\frac{y_{F_c}^{k,n}-y_{obs}^i}{a_y^i}\right)^2, \tag{55p}$$

$$\psi_{F_s,i}^{k,n} \le \left(\frac{x_{F_s}^{k,n}-x_{obs}^i}{a_x^i}\right)^2+\left(\frac{y_{F_s}^{k,n}-y_{obs}^i}{a_y^i}\right)^2, \tag{55q}$$

$$\psi_{F_0,i}^{n} \le \left(\frac{x_{F_0}^{n}-x_{obs}^i}{a_x^i}\right)^2+\left(\frac{y_{F_0}^{n}-y_{obs}^i}{a_y^i}\right)^2, \tag{55r}$$

(1), (15), (20)–(22),

where $n=1,\ldots,L+1$, $i=1,..,G$ and $k=1,\ldots,K$. Similar to [20], constraints (55i)–(55r) are obtained by replacing the equality sign with an inequality sign from corresponding constraints. Since constraints (55i)–(55r) are non-convex [23], P2.1 is a non-convex problem.

To solve P2.1, we first deal with the non-convex constraints by finding the global upper bounds or lower bounds of the non-convex terms in the non-convex constraints. For constraint (55i), its right hand side (RHS) is non-convex. According to optimization theory, the RHS satisfies the following inequity constraint [23], i.e.,

$$\begin{aligned}&\ln\left[\frac{1}{1-e^{-\tau(\|\tilde{\mathbf{q}}_k-\mathbf{a}_k\|^2+(H_q^k)^2)^{1/2}}}\right]\\&\ge \ln\left[\frac{1}{1-e^{-\tau(\|\tilde{\mathbf{q}}_k^l-\mathbf{a}_k\|^2+(H_q^{k,l})^2)^{1/2}}}\right]\\&\quad-\zeta_k(\|\tilde{\mathbf{q}}_k-\mathbf{a}_k\|^2+(H_q^k)^2-\|\tilde{\mathbf{q}}_k^l-\mathbf{a}_k\|^2-(H_q^{k,l})^2)\\&=Y_{lb}^{l,k},\end{aligned} \tag{56}$$

where l denotes an iteration number, $(\tilde{\mathbf{q}}_k^{l\,\mathrm{T}}, H_q^{k,l})$ denotes the UAV's hovering position for DC at the lth iteration, $Y_{lb}^{l,k}$ denotes the global lower bound (GLB) of the RHS at the lth iteration, and ζ_k is given by

$$\zeta_k=\frac{\tau\cdot e^{-\tau(\|\tilde{\mathbf{q}}_k^l-\mathbf{a}_k\|^2+(H_q^{k,l})^2)^{1/2}}}{2[1-e^{-\tau(\|\tilde{\mathbf{q}}_k^l-\mathbf{a}_k\|^2+(H_q^{k,l})^2)^{1/2}}](\|\tilde{\mathbf{q}}_k^l-\mathbf{a}_k\|^2+(H_q^{k,l})^2)^{1/2}}.$$

For constraint (55j), let $R_{lb}^{l,k}$ denote the GLB of its RHS at the lth iteration. Similarly, we have

$$\begin{aligned}&\log_2\left\{1+\frac{\gamma_0}{[\|\tilde{\mathbf{p}}_k\|^2+(H_p^k-H_b)^2]^{\alpha/2}}\right\}\\&\ge\log_2\left\{1+\frac{\gamma_0}{[\|\tilde{\mathbf{p}}_k^l\|^2+(H_p^{k,l}-H_b)^2]^{\alpha/2}}\right\}\\&\quad-\phi_k^l[\|\tilde{\mathbf{p}}_k\|^2+(H_p^k-H_b)^2-\|\tilde{\mathbf{p}}_k^l\|^2-(H_p^{k,l}-H_b)^2]\\&=R_{lb}^{l,k},\end{aligned} \tag{57}$$

where $(\tilde{\mathbf{p}}_k^{l\,\mathrm{T}}, H_p^{k,l})$ denotes the UAV's local hovering position for DT at the lth iteration and ϕ_k^l is given by

$$\phi_k^l=\frac{(\log_2 e)\gamma_0\alpha/2}{[\|\tilde{\mathbf{p}}_k^l\|^2+(H_p^{k,l}-H_b)^2]\{[\|\tilde{\mathbf{p}}_k^l\|^2+(H_p^{k,l}-H_b)^2]^{\alpha/2}+\gamma_0\}}.$$

For constraints (55k)–(55m), let $\breve{U}_{F_c}^{k,n,l}$, $\breve{U}_{F_s}^{k,n,l}$ and $\breve{U}_{F_0}^{n,l}$ denote the GLBs of their RHSs, respectively. Similarly, we have

$$\begin{aligned}&(\mu_{F_c}^{k,n})^2+2(\Delta_{F_c}^{k,n})^2/c_4\ge-(\mu_{F_c}^{k,n,l})^2+2\mu_{F_c}^{k,n,l}\mu_{F_c}^{k,n}\\&-2/c_4\times\Big[\Big\|\mathbf{w}_{F_c}^{k,n-1,l}-\mathbf{w}_{F_c}^{k,n,l}\Big\|^2+(z_{F_c}^{k,n-1,l}-z_{F_c}^{k,n,l})^2\\&+2(\mathbf{w}_{F_c}^{k,n-1,l}-\mathbf{w}_{F_c}^{k,n,l})^{\mathrm{T}}(\mathbf{w}_{F_c}^{k,n-1}-\mathbf{w}_{F_c}^{k,n})\\&+2(z_{F_c}^{k,n-1,l}-z_{F_c}^{k,n,l})(z_{F_c}^{k,n-1}-z_{F_c}^{k,n})\Big]=\breve{U}_{F_c}^{k,n,l},\end{aligned} \tag{58}$$

where $(\mathbf{w}_{F_c}^{k,n,l\,\mathrm{T}}, z_{F_c}^{k,n,l})$denotes the UAV's waypoint for collecting *DATA_k* at the lth iteration. Since $\breve{U}_{F_s}^{k,n,l}$ and $\breve{U}_{F_0}^{n,l}$ both have the similar form to that of $\breve{U}_{F_c}^{k,n,l}$, we will not unnecessarily give the formulas of $\breve{U}_{F_s}^{k,n,l}$ and $\breve{U}_{F_0}^{n,l}$ here.

For constraints (55n)–(55r), let $\breve{O}_{q,i}^{k,l}$, $\breve{O}_{p,i}^{k,l}$, $\breve{O}_{F_c,i}^{k,n,l}$, $\breve{O}_{F_s,i}^{k,n,l}$ and $\breve{O}_{F_0,i}^{n,l}$ denote the GLBs of their RHSs, respectively. Similarly, we have

$$\begin{aligned}&\left(\frac{x_q^k-x_{obs}^i}{a_x^i}\right)^2+\left(\frac{y_q^k-y_{obs}^i}{a_y^i}\right)^2\\&\ge 2\left(\frac{x_q^{k,l}-x_{obs}^i}{(a_x^i)^2}\right)(x_q^k-x_q^{k,l})\\&\quad+2\left(\frac{y_q^{k,l}-y_{obs}^i}{(a_y^i)^2}\right)(y_q^k-y_q^{k,l})\end{aligned}$$

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$$+\left(\frac{x_q^{k,l}-x_{obs}^i}{a_x^i}\right)^2+\left(\frac{y_q^{k,l}-y_{obs}^i}{a_y^i}\right)^2=\breve{O}_{q,i}^{k,l}, \tag{59}$$

where $(x_q^{k,l}, y_q^{k,l})^{\mathrm{T}} = \tilde{\mathbf{q}}_k^l$ denotes the UAV's hovering position for DC at the lth iteration. Since $\breve{O}_{p,i}^{k,l}$, $\breve{O}_{F_c,i}^{k,n,l}$, $\breve{O}_{F_s,i}^{k,n,l}$ and $\breve{O}_{F_0,i}^{n,l}$ all have the similar form to that of $\breve{O}_{q,i}^{k,l}$, we will not unnecessarily give the formulas of $\breve{O}_{p,i}^{k,l}$, $\breve{O}_{F_c,i}^{k,n,l}$, $\breve{O}_{F_s,i}^{k,n,l}$, and $\breve{O}_{F_0,i}^{n,l}$ here.

By replacing the RHSs of constraints (55i)–(55r) with their GLBs at the lth iteration, respectively, P2.1 can be rewritten as P2.2, i.e.,

P2.2

Objective:

$$\min_{\mathcal{Q},\mathcal{W},\mathcal{T},\mathcal{L}} \left\{\sum_{k=1}^{K}\left(\begin{array}{l}\sum_{n=1}^{L+1} t_{F_c}^{k,n}+\sum_{n=1}^{L+1} t_{F_s}^{k,n}+\frac{S_k}{B\eta_k}\\ +T_s\ln(\frac{1}{1-\omega})/\varphi_k\end{array}\right)+\sum_{n=1}^{L+1} t_{F_0}^{n}\right\}, \tag{60a}$$

subject to:

$$\eta_k \le R_{lb}^{l,k}, \tag{60b}$$

$$\varphi_k \le Y_{lb}^{l,k}, \tag{60c}$$

$$(t_{F_c}^{k,n})^4/(\mu_{F_c}^{k,n})^2 \le \breve{U}_{F_c}^{k,n,l}, \tag{60d}$$

$$(t_{F_s}^{k,n})^4/(\mu_{F_s}^{k,n})^2 \le \breve{U}_{F_s}^{k,n,l}, \tag{60e}$$

$$(t_{F_0}^{n})^4/(\mu_{F_0}^{n})^2 \le \breve{U}_{F_0}^{n,l}, \tag{60f}$$

$$\psi_{q,i}^k \le \breve{O}_{q,i}^{k,l}, \tag{60g}$$

$$\psi_{p,i}^k \le \breve{O}_{p,i}^{k,l}, \tag{60h}$$

$$\psi_{F_c,i}^{k,n} \le \breve{O}_{F_c,i}^{k,n,l}, \tag{60i}$$

$$\psi_{F_s,i}^{k,n} \le \breve{O}_{F_s,i}^{k,n,l}, \tag{60j}$$

$$\psi_{F_0,i}^{n} \le \breve{O}_{F_0,i}^{n,l}, \tag{60k}$$

(1), (15), (20)–(22), (55b)–(55h),

where $n = 1, \ldots, L+1$, $k = 1, \ldots, K$ and $i = 1, \ldots, G$. According to [23], all constraints of P2.2 are convex. Thus, P2.2 is a convex problem, which can be efficiently solved by using CVX [24].

Based on the above analyses, P2 can be solved by solving P2.1 because P2.1 is equivalent to P2. Since the optimal objective value of P2.2 actually provides a global upper bound to that of P2.1, P2.1 can be solved by using an SCA method [14], [20]. Therefore, we propose a SCA-based optimization algorithm to solve P2.1 or P2, which is described in Algorithm 2.

C. 3D Trajectory Optimization Algorithm and Computational Complexity

The 3D trajectory optimization algorithm is based on Algorithm 1 and Algorithm 2 proposed to respectively solve P1 and P2, as described in Algorithm 3. The main idea of the 3D trajectory optimization algorithm is to alternatively optimize the UAV's visiting sequence, hovering positions, and flying paths between the hovering positions until the UAV's total mission completion time converges.

Algorithm 2: SCA-Based Optimization Algorithm.

Input: the UAV's visiting sequence:
$\boldsymbol{\pi} = [\pi(1), \ldots, \pi(N)]$,
a set of the UAV's initial hovering positions:
$\mathcal{Q}^0 = \{\tilde{\mathbf{q}}_k^0, \tilde{\mathbf{p}}_k^0, H_q^{k,0}, H_p^{k,0}, k = 1, \ldots, K\}$,
a set of the UAV's initial waypoints:
$\mathcal{W}^0 = \{\mathbf{w}_{F_c}^{k,n,0}, \mathbf{w}_{F_s}^{k,n,0}, \mathbf{w}_{F_0}^{n,0}, z_{F_c}^{k,n,0}, z_{F_s}^{k,n,0}, z_{F_0}^{n,0}, k = 1, \ldots, K, n = 1, \ldots, L\}$;

1: Set $l = 0$;
2: **Repeat:** Solve P2.2 with
$\mathcal{Q}^l = \{\tilde{\mathbf{q}}_k^l, \tilde{\mathbf{p}}_k^l, H_q^{k,l}, H_p^{k,l}, k = 1, \ldots, K\}$ and
$\mathcal{W}^l = \{\mathbf{w}_{F_c}^{k,n,l}, \mathbf{w}_{F_s}^{k,n,l}, \mathbf{w}_{F_0}^{n,l}, z_{F_c}^{k,n,l}, z_{F_s}^{k,n,l}, z_{F_0}^{n,l}, k = 1, \ldots, K, n = 1, \ldots, L\}$
by using CVX to find
$\mathcal{Q}^{l+1} = \{\tilde{\mathbf{q}}_k^{l+1}, \tilde{\mathbf{p}}_k^{l+1}, H_q^{k,l+1}, H_p^{k,l+1}, k = 1, \ldots, K\}$,
$\mathcal{W}^{l+1} = \{\mathbf{w}_{F_c}^{k,n,l+1}, \mathbf{w}_{F_s}^{k,n,l+1}, \mathbf{w}_{F_0}^{n,l+1}, z_{F_c}^{k,n,l+1}, z_{F_s}^{k,n,l+1}, z_{F_0}^{n,l+1}, k = 1, \ldots, K, n = 1, \ldots, L\}$
and obtain T_{total}^{l+1};
3: Perform $l = l + 1$;
4: **Until** the objective value T_{total} converges;

Output: $\mathcal{Q}^* = \{\tilde{\mathbf{q}}_k*, \tilde{\mathbf{p}}_k*, H_q^{k*}, H_p^{k*}, k = 1, \ldots, K)$,
$\mathcal{W}^* = \{\mathbf{w}_{F_c}^{k,n*}, \mathbf{w}_{F_s}^{k,n*}, \mathbf{w}_{F_0}^{n*}, z_{F_c}^{k,n*}, z_{F_s}^{k,n*}, z_{F_0}^{n*}, k = 1, \ldots, K, n = 1, \ldots, L\}$,
$\mathcal{T}^* = \{t_{F_s}^{k,n*}, t_{F_s}^{k,n*}, t_{F_0}^{n*}, k = 1, \ldots, K, n = 1, \ldots, L+1\}$.

The computational complexity of the 3D trajectory optimization algorithm mainly consists of two components: the complexity of Algorithm 1 and that of Algorithm 2. Similar to [14], the complexity of Algorithm 1 is approximately $O(YZK)$ and that of Algorithm 2 is approximately $O((2KGL + 2KG + GL + 10KL + 12K + 5L + 2)^{3.5}\log(1/\varepsilon))$, where Y and Z denote the maximum number of iterations and the initial population size used in Algorithm 1, respectively, $(2KGL + 2KG + GL + 10KL + 12K + 5L + 2)$ denotes the total number of the optimization variables in P2.2, and ε denotes a solution accuracy [14]. Therefore, the total computational complexity of the 3D trajectory optimization algorithm is approximately$O(L_3(YZK + (10KL + 2KGL)^{3.5}\log(1/\varepsilon)))$, where L_3 denotes the number of iterations in the algorithm.

V. Simulation Results

We show simulation results to evaluate the performance of the proposed 3D trajectory optimization algorithm. We use MATLAB to conduct the simulation experiments, and use the parameter values given in Table I [4], [14], [20] and Table II [5] in the experiments.

For evaluation, a 2D avoidance trajectory optimization (AoI-EaTO) algorithm proposed in [14] is used as a benchmark algorithm. Unlike the proposed 3D trajectory optimization algorithm, the AoI-EaTO algorithm assumes that the UAV flies at a constant altitude. In this case, only when the UAV flies at

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Algorithm 3: 3D Trajectory Optimization Algorithm.

Input: a set of the UAV's initial hovering positions: $\mathcal{Q}^0 = \{\tilde{\mathbf{q}}_k^0, \tilde{\mathbf{p}}_k^0, H_q^{k,0}, H_p^{k,0}, k = 1, \ldots, K\}$;
a set of the UAV's initial waypoints: $\mathcal{W}^0 = \{\mathbf{w}_{F_c}^{k,n,0}, \mathbf{w}_{F_s}^{k,n,0}, \mathbf{w}_{F_0}^{n,0}, z_{F_c}^{k,n,0}, z_{F_s}^{k,n,0}, z_{F_0}^{n,0}, k = 1, \ldots, K, n = 1, \ldots, L\}$;

1: Set $l = 0$;
2: **Repeat**: Solve P1 with $\mathcal{Q}^l = \{\tilde{\mathbf{q}}_k^l, \tilde{\mathbf{p}}_k^l, H_q^{k,l}, H_p^{k,l}, k = 1, \ldots, K\}$ and $\mathcal{W}^l = \{\mathbf{w}_{F_c}^{k,n,l}, \mathbf{w}_{F_s}^{k,n,l}, \mathbf{w}_{F_0}^{n,l}, z_{F_c}^{k,n,l}, z_{F_s}^{k,n,l}, z_{F_0}^{n,l}, k = 1, \ldots, K, n = 1, \ldots, L\}$ by using Algorithm 1 to find $\boldsymbol{\pi}^{l+1} = [\pi^{l+1}(1), \ldots, \pi^{l+1}(N)]$;
3: Solve P2 with $\boldsymbol{\pi}^{l+1} = [\pi^{l+1}(1), \ldots, \pi^{l+1}(N)]$, $\mathcal{Q}^l = \{\tilde{\mathbf{q}}_k^l, \tilde{\mathbf{p}}_k^l, H_q^{k,l}, H_p^{k,l}, k = 1, \ldots, K\}$ and $\mathcal{W}^l = \{\mathbf{w}_{F_c}^{k,n,l}, \mathbf{w}_{F_s}^{k,n,l}, \mathbf{w}_{F_0}^{n,l}, z_{F_c}^{k,n,l}, z_{F_s}^{k,n,l}, z_{F_0}^{n,l}, k = 1, \ldots, K, n = 1, \ldots, L\}$ by using Algorithm 2 to find $\mathcal{Q}^{l+1} = \{\tilde{\mathbf{q}}_k^{l+1}, \tilde{\mathbf{p}}_k^{l+1}, H_q^{k,l+1}, H_p^{k,l+1}, k = 1, \ldots, K\}$, $\mathcal{W}^{l+1} = \{\mathbf{w}_{F_c}^{k,n,l+1}, \mathbf{w}_{F_s}^{k,n,l+1}, \mathbf{w}_{F_0}^{n,l+1}, z_{F_c}^{k,n,l+1}, z_{F_s}^{k,n,l+1}, z_{F_0}^{n,l+1}, k = 1, \ldots, K, n = 1, \ldots, L\}$ and obtain T_{total}^{l+1};
4: Perform $l = l + 1$;
5: **Until** the objective value T_{total} converges;

Output: $\boldsymbol{\pi}^* = [\pi^*(1), \ldots, \pi^*(N)]$, $\mathcal{Q}^* = \{\tilde{\mathbf{q}}_k*, \tilde{\mathbf{p}}_k*, H_q^{k*}, H_p^{k*}, k = 1, \ldots, K)$, $\mathcal{W}^* = \{\mathbf{w}_{F_c}^{k,n*}, \mathbf{w}_{F_s}^{k,n*}, \mathbf{w}_{F_0}^{n*}, z_{F_c}^{k,n*}, z_{F_s}^{k,n*}, z_{F_0}^{n*}, k = 1, \ldots, K, n = 1, \ldots, L\}$, $\mathcal{T}^* = \{t_{F_s}^{k,n*}, t_{F_s}^{k,n*}, t_{F_0}^{n}{}^*, k = 1, \ldots, K, n = 1, \ldots, L + 1\}$.

TABLE I
SIMULATION PARAMETERS

Parameters	Values	Parameters	Values
K	12	L	50
G	12	H_0	100 m
H_b	20 m	θ_s	15°
r_a	10 m	τ	0.01
ω	99%	T_s	1s
S_k	(20–120) Mbits	B	1 MHz
σ^2	−110 dBm	p_u	23 dBm
β_0	−60 dB	κ	0.2
Γ	8.2 dB	g_1	10
g_2	0.6	α	2.2
$\overline{P}_{LoS}$	67.9%	$\gamma_{\min}$	5 dB
$v_{\max}$	40 m/s	P_c	10 W
P_s	5 W	c_1	79.86 W
c_2	0.00021	c_3	88.63 W
c_4	32.48	c_5	0.0092
T_{limit}^k	(36-45.5) s	E_{limit}	(250-340) KJ

TABLE II
OBSTACLE MODEL PARAMETERS

No. i	h_{obs}^i / m	x_{obs}^i / m	y_{obs}^i / m	x_s^i / m	y_s^i / m
1	61.6	−150	−750	80	80
2	89.2	470	−160	120	120
3	59.3	−160	440	150	150
4	93.7	−640	−370	100	150
5	85.4	740	820	80	100
6	57	40	690	100	100
7	76	−660	820	100	100
8	48	740	470	150	150
9	36	80	−380	120	120
10	79	660	−740	80	80
11	81.2	300	290	80	80
12	58.7	−620	310	120	100

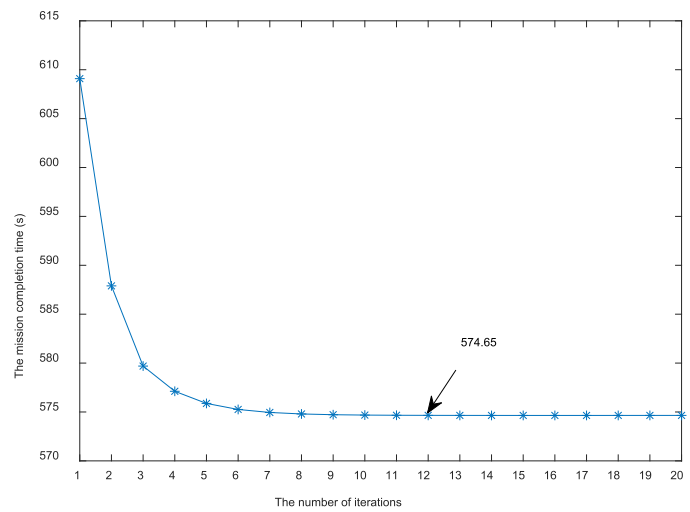

Fig. 5. The iteration process of the proposed algorithm in situation I.

an altitude high enough can it avoid collisions with all obstacles. Moreover, we introduce the following metrics: the UAV's trajectory, total mission completion time, average flying speed, total flying distance, total hovering time and total flying time.

Fig. 5 shows the UAV's total mission completion time (T_{total}) vs the number of iterations with the proposed algorithm in a situation (Situation I), where $T_{limit}^k = 45$ s, $k = 1, \ldots, K$ and $E_{limit} = 280$ KJ. It is seen that T_{total} quickly decreases and tends to stabilize within a few iterations, which shows that the proposed algorithm can converge quickly.

Figs. 6 and 7 show the UAV's trajectories with the proposed algorithm and the AoI-EaTO algorithm, respectively, in Situation I. Table III shows the other performance metrics with the two algorithms, respectively. It can be seen that the UAV's minimum total mission completion time ($T_{total}{}^*$) with the proposed algorithm is smaller than that with the AoI-EaTO algorithm. The reason is that with the proposed algorithm the UAV can hover at a lower altitude to collect or transmit data compared with the AoI-EaTO algorithm. Based on the analyses in Section III-B, a lower altitude for collecting or transmitting data always leads to a shorter DC time or DT time for the UAV. Thus, the UAV's total hovering time with the proposed algorithm is smaller than that with the AoI-EaTO algorithm, resulting a smaller UAV's

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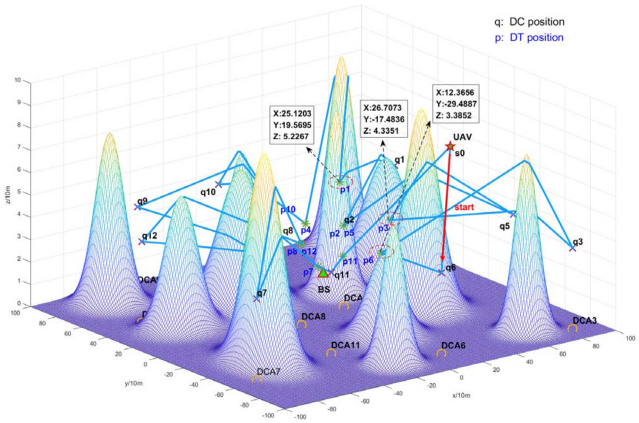

Fig. 6. The UAV's trajectory with the proposed algorithm in situation I.

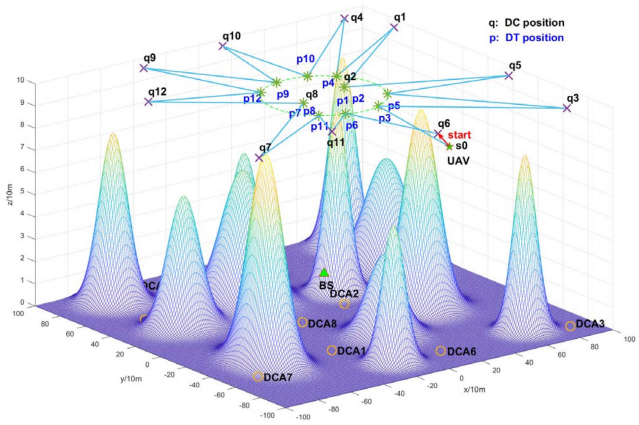

Fig. 7. The UAV's trajectory with the AoI-EaTO algorithm in situation I.

TABLE III
PERFORMANCE COMPARISON

Performance Metrics	The proposed algorithm	The AoI-EaTO algorithm
The total mission completion time	574.65 s	653.36 s
The average flying speed	38.55 m/s	37.84 m/s
The total flying distance	14569 m	14439 m
The total hovering time	196.72 s	271.72 s
The total flying time	377.93 s	381.64 s

energy consumption for hovering with the proposed algorithm. As a result, with the proposed algorithm, the UAV has more energy for flying and thus the UAV's average flying speed is larger than that with the AoI-EaTO algorithm. Moreover, the UAV's total flying distance with the proposed algorithm is larger than that with the AoI-EaTO algorithm. The reason is with the proposed algorithm the UAV needs to maneuver around or fly over obstacles to avoid collisions, resulting the longer flying distance for the UAV compared with that with the AoI-EaTO algorithm.

Figs. 8 and 9 show the UAV's minimum total mission completion time (${T_{total}}^*$) and average flying speed (v_{avg}) with the proposed algorithm under different energy constraints (E_{limit}), respectively, where $T^k_{limit} = 45$ s,$k = 1, \ldots, K$. It can be observed that an increase in E_{limit} causes ${T_{total}}^*$ to decrease and

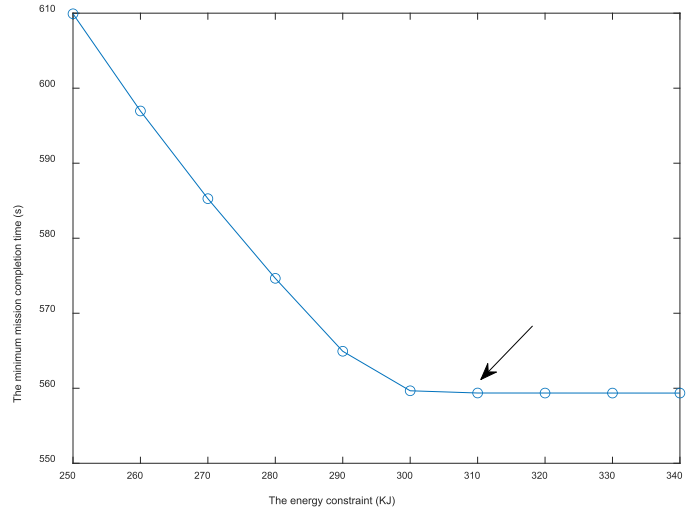

Fig. 8. ${T_{total}}^*$ vs E_{limit}.

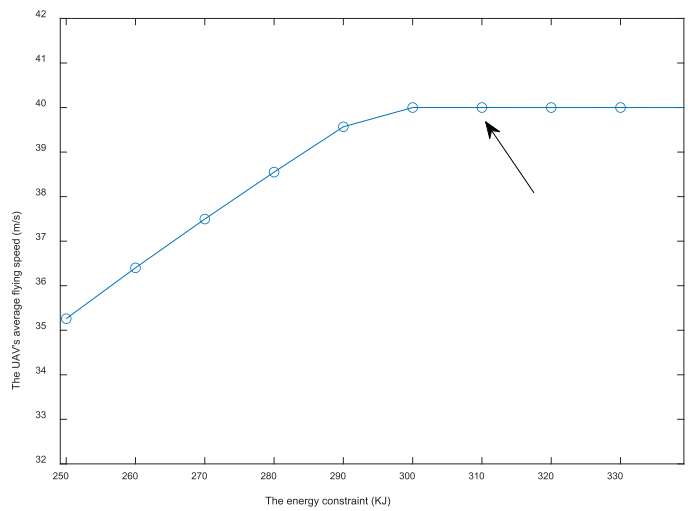

Fig. 9. v_{avg} vs E_{limit}.

subsequently stabilize. Meanwhile, v_{avg} increases and tends to $v_{\max}$. The reason is that as E_{limit} increases, the UAV can obtain more energy to fly at a faster speed, which would in turn lead to a corresponding decrease in ${T_{total}}^*$. However, if E_{limit} exceeds a certain value (e.g., 310KJ in Fig. 8), the UAV would fly at $v_{\max}$, resulting in ${T_{total}}^*$ to stabilize.

Figs. 10 and 11 show the UAV's minimum total mission completion time (${T_{total}}^*$) and average flying speed (v_{avg}) with the proposed algorithm under different time constraints of collected data (T^k_{limit}), respectively, where $E_{limit} = 280$KJ. It is seen that as T^k_{limit} increases, ${T_{total}}^*$ first decreases and subsequently stabilizes, and meanwhile v_{avg} first increases and then tends to stabilize. The reason is that when T^k_{limit} increases, the UAV would have more time to transmit *DATA_k*. In this case, the UAV would not have to fly to a position closer to the GBS to reduce the age of *DATA_k*, resulting in a decrease in the UAV's total flying distance and thus an increase in v_{avg}given the energy constraint. Consequently, ${T_{total}}^*$ would decrease. But when T^k_{limit} exceeds a certain value (e.g., 39s in Fig. 10), the UAV's total flying

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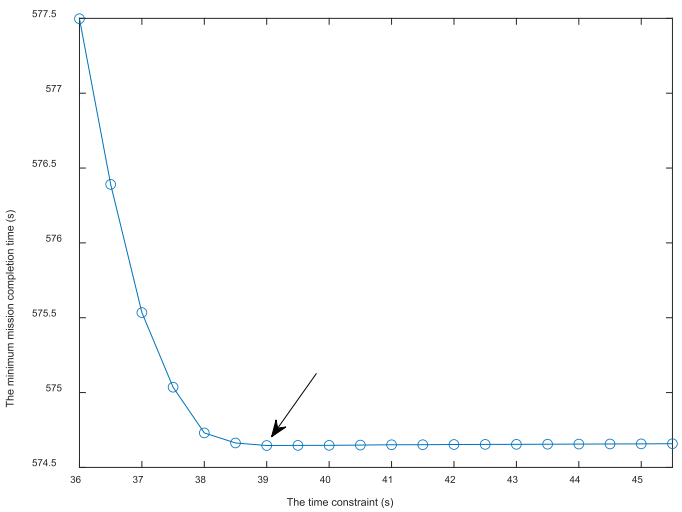

Fig. 10. $T_{total}{}^*$ vs T_{limit}^k.

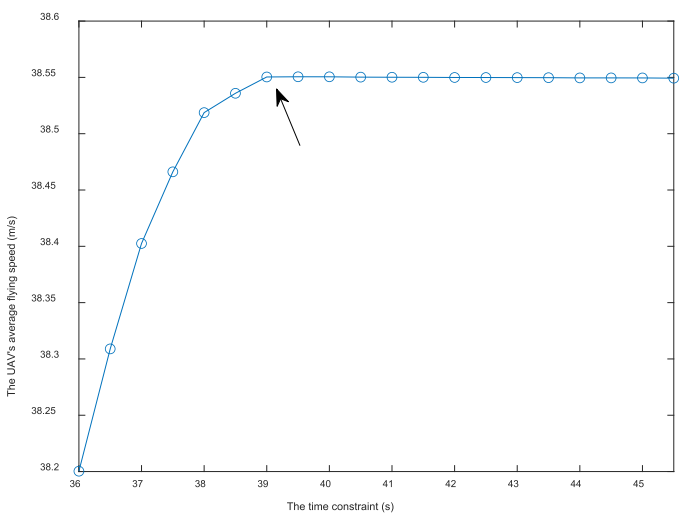

Fig. 11. v_{avg} vs T_{limit}^k.

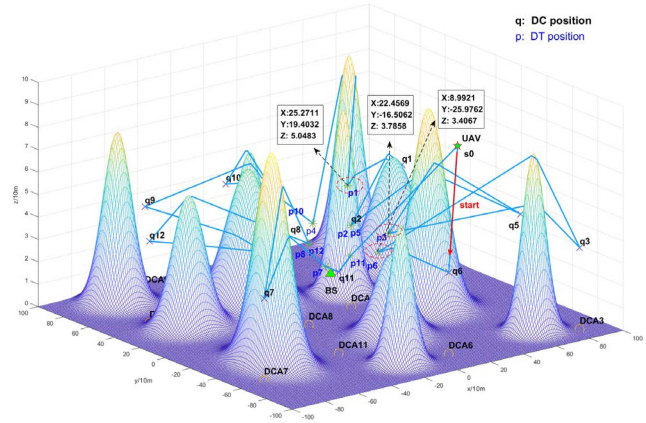

Fig. 12. The UAV's trajectory in situation II.

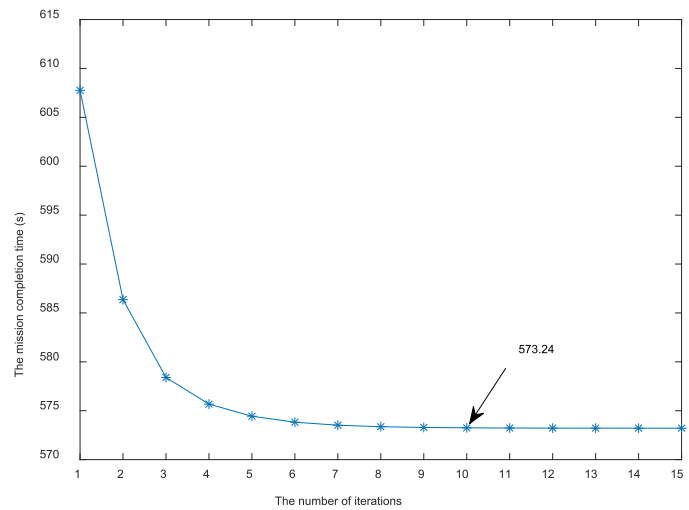

Fig. 13. The iteration process of the proposed algorithm in situation III.

distance cannot be further reduced, resulting inv_{avg} and $T_{total}{}^*$ to stabilize.

Fig. 12 shows the UAV's trajectory with the proposed algorithm in a situation (Situation II), where the time constraints of data collected in a_1, a_3, and a_6 are set to smaller values, compared with Situation I, i.e.,$T_{limit}^1 = 26$ s,$T_{limit}^3 = 36$ s,$T_{limit}^6 = 35$ s, to investigate the impact of T_{limit}^kon the UAV's trajectory. Comparing Fig. 12 with Fig. 6, the UAV's hovering positions for transmitting data collected in a_1, a_3 and a_6 in Situation II are closer to the GBS than those in Situation I. This is because in Situation II, the time constraints of data collected in a_1, a_3 and a_6 are smaller than those in Situation I. Thus, the UAV needs to fly faster and select hovering positions closer to the BS for transmitting collected data in order to meet the time constraints.

Figs. 13 and 14 show the UAV's total mission completion time (T_{total}) vs the number of iterations, and the UAV's trajectory with the proposed algorithm in a situation (called Situation III), where the parameters of the obstacle model are set differently from those in Situation I. It is seen that similar to Situation I,T_{total} also quickly decreases and tends to stabilize within a few iterations, and the UAV can avoid collisions with spatial obstacles during the overall mission as well, which further demonstrates the efficiency of the proposed algorithm.

Fig. 15 shows the UAV's total flying distance with the proposed algorithm vs the packet size of collected data (S_k), where $T_{limit}^k = 50$ s, $k = 1, \ldots, K$ and $E_{limit} = 300$ KJ. It is seen that as S_k increases, the UAV's total flying distance first remains unchanged, then gradually increases, and tends to stabilize. The reason is that as S_k increases, if the UAV's hovering positions for DT remains unchanged, the UAV's DT time would increase, which would result in the increasing of the age of collected data. When S_k is smaller than a certain value (i.e., 40Mbits), the age of collected data is within the time constraint of collected data and thus the UAV's hovering positions for DT can remain unchanged. As a result, the UAV's total flying distance remains unchanged. When S_k becomes larger, to meet the time constraint of collected data, the UAV's hovering positions for DT would be closer to the BS to reduce the age of collected data. Thus, the UAV's total flying distance would increase. However, when S_k increases to a certain value (i.e., 90Mbits), the UAV's hovering positions for DT have already been closest to the BS. As a result,

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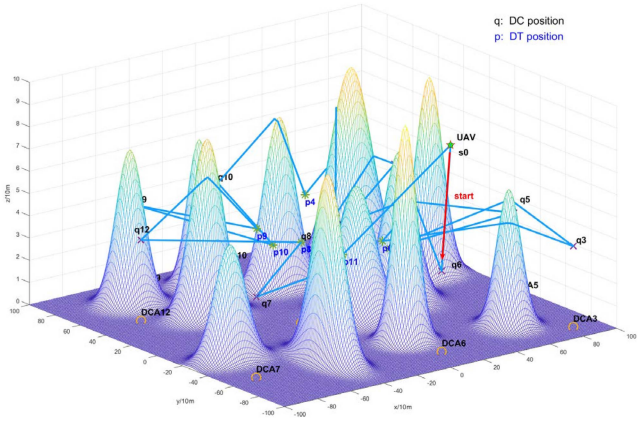

Fig. 14. The UAV's trajectory in situation III.

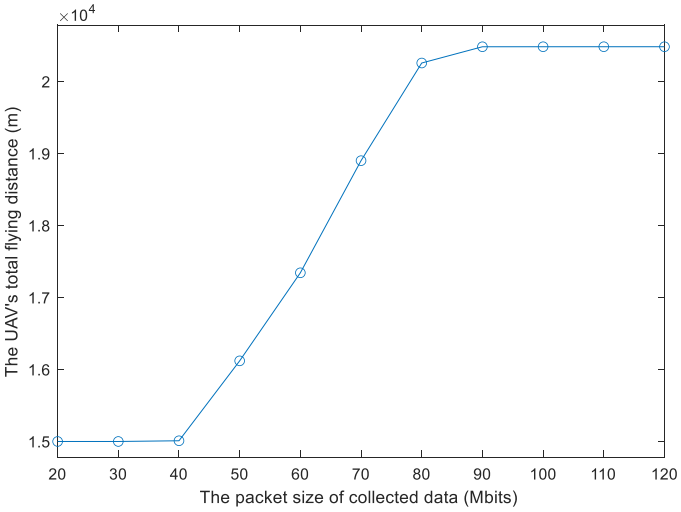

Fig. 15. The UAV's total flying distance vs S_k.

the age of collected data cannot further be reduced and thus the UAV's total flying distance would not continue to change as S_k further increases.

Fig. 16 shows the UAV's total flying time with the proposed algorithm vs the packet size of collected data (S_k), where $T_{limit}^k = 50$ s, $k = 1, \ldots, K$ and $E_{limit} = 300$ KJ. It is seen that the UAV's total flying time keeps increasing, but increases slowly when S_k is small or large. The reason is that as S_k increases, no matter whether the UAV's DT time increases or the UAV's hovering positions for DT are closer to the BS, the UAV's energy consumption for flying decreases, which would reduce the UAV's average flying speed. Since the UAV's total flying time mainly depends on the UAV's average flying speed and total flying distance, the UAV's total flying time keeps increasing with the reduction of the UAV's average flying speed and the increase of the UAV's total flying distance. According to Fig. 15, when S_k is small or large, the UAV's total flying distance remains unchanged. In this case, the UAV's total flying time only increase with the reduction of the UAV's average flying speed, thus the UAV's total flying time increases slowly.

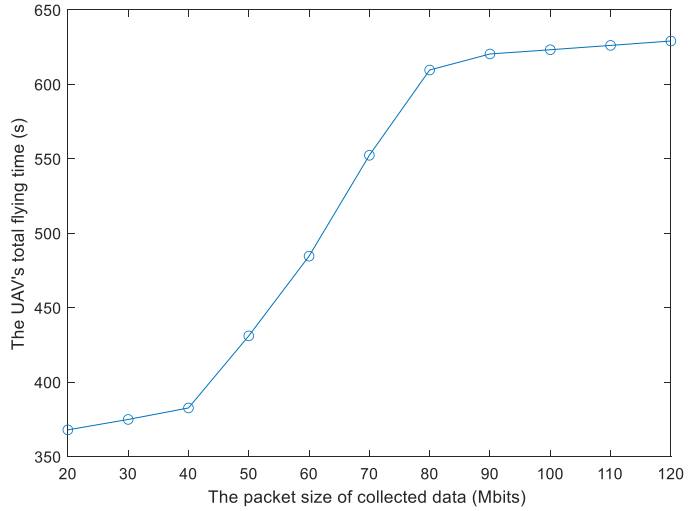

Fig. 16. The UAV's total flying time vs S_k.

VI. Conclusion and Future Work

In this paper, we investigated the 3D UAV trajectory-planning problem in a UAV-enabled DC system, in which a UAV is deployed to execute a DC mission to collect time-constrained data in a number of DCAs in a geographical area with various spatial obstacles. The problem was formulated as an optimization problem with an objective to minimize the UAV's total mission completion time, taking into account the time constraint of collected data, the avoidance of the obstacles, and the UAV's energy constraint. To solve the formulated problem, the problem was decomposed into a visiting sequence optimization sub-problem, and a hovering position and flying path optimization sub-problem. Correspondingly, a GA based algorithm, and a SCA based algorithm were proposed to solve the two sub-problems, respectively. Based on the two algorithms, a 3D trajectory optimization algorithm was further proposed to solve the main problem. The simulation results show that the proposed 3D trajectory optimization algorithm outperforms the 2D benchmark algorithm in terms of the UAV's total mission completion time, and the UAV's trajectory planned with the proposed algorithm can avoid collisions with obstacles and meet the time constraints of collected data.

In this paper, we considered a DC/DT mode in which a UAV performs DC/DT while hovering and used an energy consumption model which results in a UAV to fly over, not around obstacles. For future work, it is interesting to investigate a DC/DT mode in which a UAV performs DC/DT while flying in the planning of a 3D trajectory, and investigate whether flying around obstacles is better than flying over obstacles in energy saving for a whole 3D trajectory.

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